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High-Efficiency Double-Stranded cDNA Synthesis from dsRNA Templates

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Abdonaser Poursalavati[1,2]

[1]Department of Biology, Université de Sherbrooke;
[2]Saint-Jean-sur-Richelieu Research and Development Centre, Agriculture and Agri-Food Canada

Abdonaser Poursalavati: PhD Candidate in Soil Virology;

Abdonaser Poursalavati
Université de Sherbrooke

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We use this protocol and it's working

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Abstract

High-Efficiency Double-Stranded cDNA Synthesis from dsRNA Templates
A cost-effective, custom protocol for preparing viral dsRNA for next-generation sequencing using SuperScript IV.

This protocol details a robust and cost-effective method for synthesizing double-stranded cDNA (ds-cDNA) from dsRNA templates. It is optimized for low-input dsRNA, such as that extracted from complex environmental samples (e.g., soil viromes). The protocol uses a two-step process, starting with first-strand synthesis using SuperScript IV Reverse Transcriptase, followed by second-strand synthesis using Klenow Polymerase. The resulting ds-cDNA is suitable for library preparation and next-generation sequencing. This method serves as a reliable alternative to commercial kits.

This protocol is a key downstream component of our **SS-VIME** (Single-Source Virome-Microbiome Extraction) workflow, recently accepted for publication in ***Microbiology Spectrum***. After using the primary **SS-VIME protocol** to co-extract genomic DNA and dsRNA from a single environmental sample, this method is specifically designed to process the purified dsRNA fraction, enabling the characterization of the active RNA virome.

To facilitate high-throughput use, an **Excel-based reagent calculator** is provided in the "Attachments" section. Users simply input the number of samples, and the spreadsheet automatically calculates the required master mix volumes, including a built-in surplus to account for pipetting inaccuracies.

Background
While many commercial kits are available, they can be costly, especially for large-scale projects. This protocol was developed and optimized to provide a high-yield, cost-effective, and reliable workflow for generating high-quality ds-cDNA. It has been successfully used in our lab to prepare sequencing libraries from dsRNA isolated from challenging soil matrices, enabling the study of the active soil virome.

*Note: The associated manuscript, titled **"SS-VIME: A Single-Source Virome-Microbiome Extraction Protocol Toward Comprehensive Soil Community Analysis,"** is currently in press at Microbiology Spectrum. Until the official manuscript DOI is available, please cite this protocol using its protocols.io **DOI**.*

Attachments

XLSX

Double-Stranded cDNA...

13KB

Guidelines

Using the Excel Reagent Calculator

1. **Open the file** and locate the **blue box** labeled "Number of samples?".
2. **Enter the number of samples** you are processing.
3. The spreadsheet will automatically calculate the total volumes needed for each master mix in the corresponding **green boxes**. These calculations include a surplus for one extra reaction (n+0.5) to prevent shortages due to pipetting error.
4. Use the volumes from the **green boxes** to prepare your master mixes.
5. The **purple boxes** indicate the volume of master mix to add to each individual sample.

Materials

Enzymes & Buffers:

- SuperScript™ IV Reverse Transcriptase (Thermo Fisher Scientific)
- Inhibitor, Murine RNase
- Klenow Fragment
- E. coli DNA Ligase
- RNase H
- Random Hexamers (50 µM)
- dNTP Mix (10 mM)
- Dithiothreitol (DTT, 100 mM)
- Nuclease-Free Water

General Labware & Equipment:

- Nuclease-free 0.2 mL PCR tubes or plates
- Pipettes and nuclease-free filter tips
- Thermocycler
- Magnetic rack for bead cleanup
- SPRI beads (e.g., AMPure XP)
- Nuclease-free microfuge tubes

Troubleshooting

Before start

Using the Excel Reagent Calculator

For efficient and accurate reagent preparation, please use the attached Excel file

1. **Open the file** and locate the **blue box** labeled "Number of samples?".
2. **Enter the number of samples** you are processing.
3. The spreadsheet will automatically calculate the total volumes needed for each master mix in the corresponding **green boxes**. These calculations include a surplus for one extra reaction (n+0.5) to prevent shortages due to pipetting error.
4. Use the volumes from the **green boxes** to prepare your master mixes.
5. The **purple boxes** indicate the volume of master mix to add to each individual sample.

Before You Start

1
- Thaw all reagents on ice. Briefly centrifuge all tubes before opening to collect contents at the bottom.
- Keep enzymes on ice or in a cold block at all times.
- When preparing master mixes, calculate the required volume for your number of samples (n) plus at least one extra reaction to account for pipetting error (i.e., calculate for n+0.5). This is reflected in the Excel calculator's "err." column.

Part A: First-Strand cDNA Synthesis

1.1

Step 1: Prepare Denaturation Mix

- In a nuclease-free tube, prepare the following master mix. The volumes below are for a single reaction. Multiply by **n+0.5** for your master mix.

	A	B
	Component	Volume per reaction (µL)
	Random hexamers (50 µM)	1
	dNTPs (10 mM)	1
	Nuclease-free water	1
	Add to each sample (µL)	**3**

1.2 **Step 2: Set up Denaturation Reaction**

- In individual nuclease-free PCR tubes, add **10 µL** of your dsRNA template.
- Add **3 µL** of First-Strand Master Mix #1 to each tube. The total volume should be 13 µL.
- Gently mix by pipetting and centrifuge briefly.

1.3 **Step 3: Incubation - Denaturation**

- Place the tubes in a thermocycler and run the following program:
  - 5 minutes at 95°C

- Hold at 4°C
- Immediately after the program finishes, place the tubes on ice for at least 1 minute.

1.4 **Step 4: Prepare First-Strand Master Mix (RT Mix)**

While the denaturation is running, prepare the following master mix. The volumes below are for a single reaction. Multiply by **n+0.5**.

	A	B
	Component	Volume per reaction (µL)
	SuperScript IV Buffer (5X)	4
	DTT (100 mM)	1
	Inhibitor, Murine RNase (40 U/µL)	0.5
	SuperScript IV RT (200 U/µL)	1
	Nuclease-free water	0.5
	Add to each sample (µL)	**7**

1.5 **Step 5: First-Strand Synthesis**

- Add 7 µL of First-Strand Master Mix #2 to each of your 13 µL reactions on ice. The total volume is now 20 µL.
- Mix gently by pipetting and centrifuge briefly.

1.6 **Step 6: Incubation - First-Strand Synthesis**

- Return the tubes to the thermocycler and run the following program:
  - 10 minutes at 23°C (for primer annealing)
  - 30 minutes at 55°C (for reverse transcription)
  - 10 minutes at 80°C (to inactivate the enzyme)
  - Hold at 4°C
- After the program finishes, place the tubes on ice for at least 1 minute. You can proceed to the next step or store the samples at -20°C.

Part B: Second-Strand cDNA Synthesis

2 **Step 7: Prepare Second-Strand Master Mix**

2.1
- In a nuclease-free tube, prepare the following master mix. The volumes below are for a single reaction. Multiply by n+0.5.
- Pro-Tip: The RNase H volume is very small (0.2 µL). When preparing the master mix, pipette this component carefully and ensure the mix is thoroughly vortexed before aliquoting.

	A	B
	Component	Volume per reaction (µL)
	Klenow 10X buffer (+NAD)	2.5
	dNTP (10 mM)	1
	Klenow polymerase (5 U/µL)	0.5
	E.coli Ligase I (10 U/µL)	1
	Rnase H (5 U/µL)	0.2
	Add to each sample (µL)	**5.2**

2.2 **Step 8: Second-Strand Synthesis**

- To each 20 µL first-strand reaction, add 5.2 µL of Second-Strand Master Mix #3. The final volume will be 25.2 µL.
- Mix gently by pipetting and centrifuge briefly.

2.3 **Step 9: Incubation - Second-Strand Synthesis**

- Return the tubes to the thermocycler and run the following program:

  - 150 minutes (2.5 hours) at 16°C
  - 20 minutes at 75°C (to inactivate enzymes)
  - Hold at 4°C

3

Part C: ds-cDNA Cleanup

4 **Step 10: Bead Purification**

- Perform a standard SPRI bead cleanup to purify the final double-stranded cDNA.

- Use a bead-to-sample ratio of **1.8X**. For a ~25 µL sample, this corresponds to **45 µL** of beads.
- Follow the bead manufacturer's instructions for binding, washing (with 80% ethanol), and elution.
- Elute the purified ds-cDNA in an appropriate volume (e.g., 20-30 µL) of nuclease-free water or buffer. The cDNA is now ready for quantification and library preparation.

Protocol references

SS-VIME: Single-Source Virome-Microbiome Extraction Protocol For simultaneous profiling of viral, bacterial, and fungal communities from a single soil sample dx.doi.org/10.17504/protocols.io.kxygxp64zl8j/v1